

Question 4

1. $c(t) = A_c \cos(2\pi f_c t)$ is modulated by a single ~~tone~~ tone
 $m(t) = A_m \cos(2\pi f_m t)$ to obtain

$$s(t) = [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t).$$

$$A_c = 10,$$

$$A_m = 2,$$

$$f_c = 1000 \text{ Hz}.$$

$$f_m = 50 \text{ Hz}.$$

- (a) Rewrite $s(t)$ to obtain the sidebands of $s(t)$ in time domain.

$$\begin{aligned} s(t) &= A_c \cos 2\pi f_c t + A_m \cos(2\pi f_m t) \cdot \cos 2\pi f_c t \\ &= 10 \cos(2\pi \times 1000 t) + 2 \cos(2\pi \times 50 t) \cdot \cos(2\pi \times 1000 t) \\ &= 10 \cos(2000\pi t) + 2 \cos(100\pi t) \cdot \cos(2000\pi t). \end{aligned}$$

$$\boxed{2 \cos A \cos B = \cos(A+B) + \cos(A-B)}$$

$$s(t) = 10 \cos(2000\pi t) + \frac{2}{2} (\cos(2100\pi t) + \cos(1900\pi t))$$

or
$$= 10 \cos(2000\pi t) + \cos(2100\pi t) + \cos(1900\pi t)$$

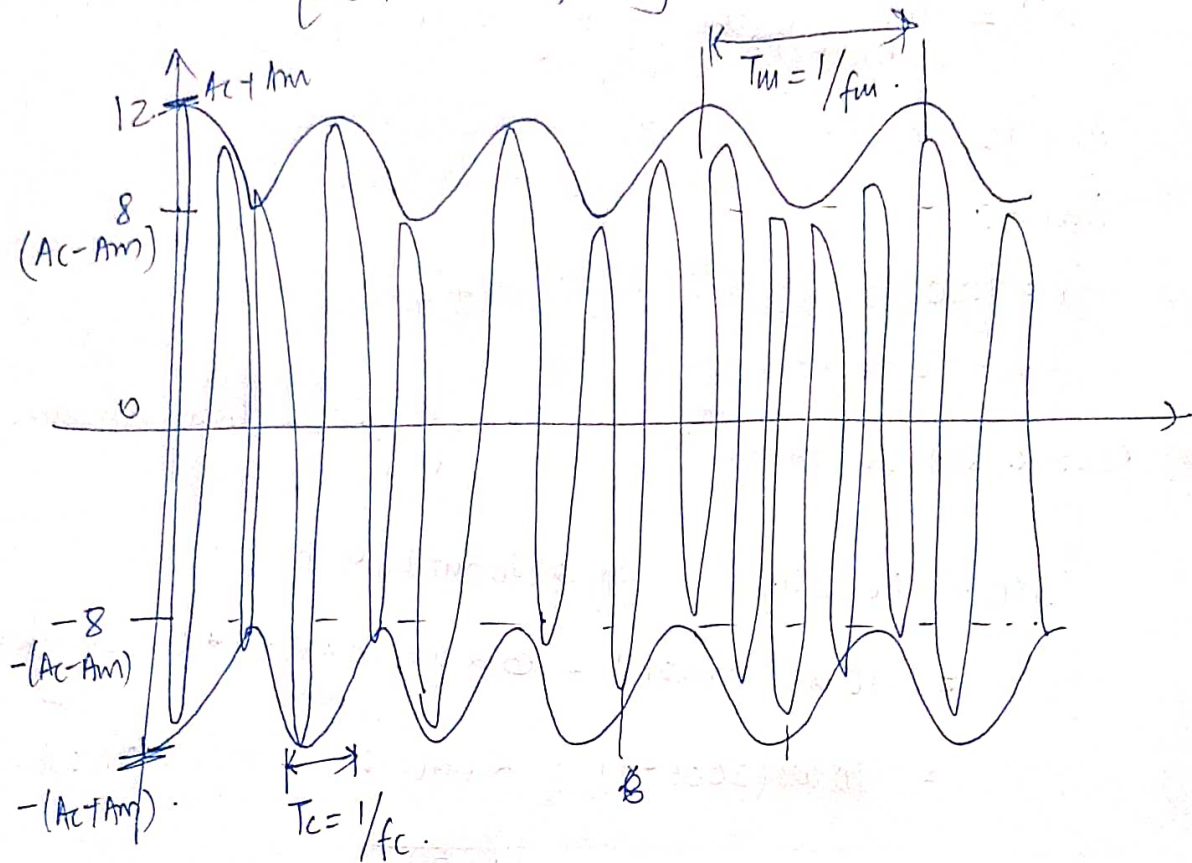
$$s(t) = A_c \cos 2\pi f_c t + \frac{A_m}{2} \left(\cos(2\pi (f_c - f_m) t) + \cos(2\pi (f_c + f_m) t) \right).$$

(b) Sketch SSB. Depict the message signal, carrier waveform & modulated amplitude level.

$$\mu = \frac{A_m}{A_c} \Rightarrow \mu = \frac{2}{10} = \underline{0.2}$$

Hence message appears on envelope.

$$SSB = [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t).$$

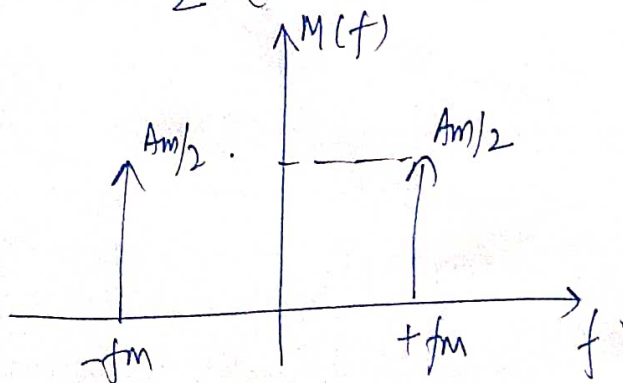


(c) Sketch the freq. spectra.

$$|M(f)|, |C(f)|, |S(f)|.$$

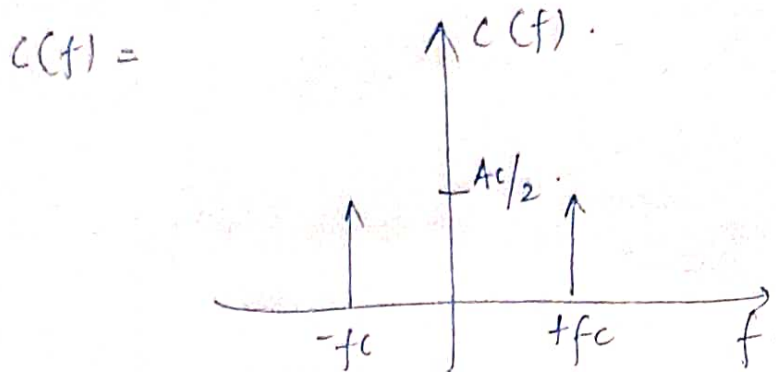
$$m(t) = A_m \cos(2\pi f_m t)$$

$$|M(f)| = \frac{A_m}{2} (\delta(f - f_m) + \delta(f + f_m)).$$



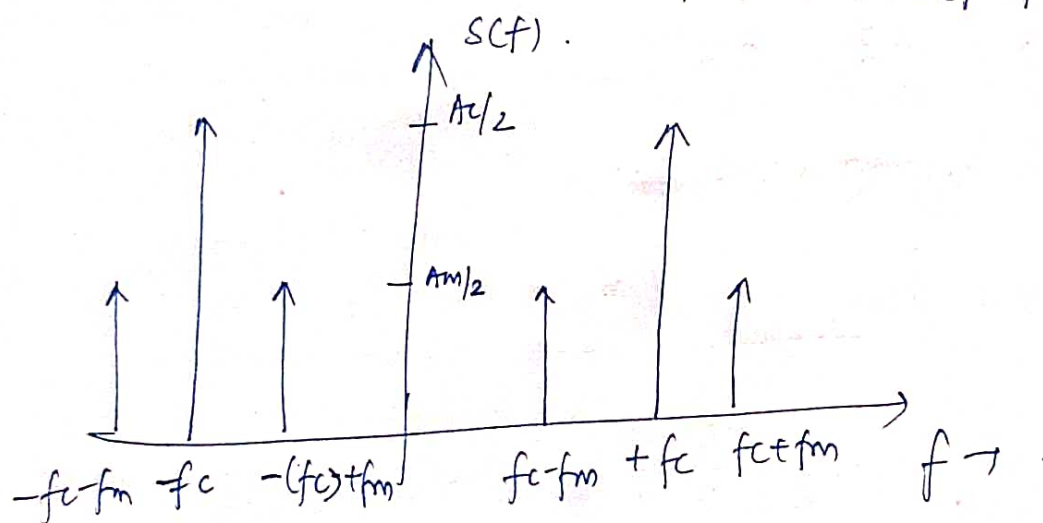
$$c(t) = A_c \cos(2\pi f_c t)$$

$$|C(f)| = \frac{A_c}{2} [\delta(f-f_c) + \delta(f+f_c)]$$



$$s(t) = (A_c + A_m \cos 2\pi f_m t) \cdot \cos(2\pi f_c t)$$

$$S(f) = \frac{A_c}{2} [\delta(f+f_c) + \delta(f-f_c)] + \frac{A_m}{4} [\delta(f-f_c-f_m) + \delta(f-f_c+f_m) + \delta(f+f_c-f_m) + \delta(f+f_c+f_m)]$$



(d) Modulation Index.

μ

$$\mu = \frac{A_m}{A_c} = \frac{2}{10} = \underline{\underline{0.2}}$$